



University of Waterloo

Department of Mechanical and Mechatronics Engineering

Design and Analysis of a Thrust Structure for a Hybrid Sounding Rocket

A report prepared for:

Waterloo Rocketry

Waterloo, ON, Canada

Shirley Kong

4A Mechanical Engineering

20 September 2021

8-250 Keats Way
Waterloo, ON, Canada
N2L 6J5

10 Sep, 2021

Professor Mike Collins
Associate Professor
Department of Mechanical and Mechatronics Engineering
University of Waterloo
Waterloo, Ontario
N2L 3G1

Dear Professor Collins,

This report, titled “Design and Analysis of a Thrust Structure for a Hybrid Sounding Rocket”, was prepared for the Waterloo Rocketry team as my 4A Work Term Report. Waterloo Rocketry is a student design team that researches, designs, builds, and launches hybrid rockets for an international competition, the Spaceport America Cup (SAC), thereby providing opportunities for students to gain hands-on experience and tackle interesting rocket-related engineering problems.

As a member of the student design team, my role on the airframe subteam involves structural analysis and fabrication of composite parts, as well as other tasks such as outreach, finances, and test operations.

The purpose of this report is to document the design process for a section of the rocket, as well as areas of future improvement. This report was written entirely by me and has not received any previous academic credit at this or any other institution.

Sincerely Yours,

A handwritten signature in black ink, appearing to read 'Shirley Kong', with a stylized, flowing script.

Shirley Kong

Contents

1	Introduction	1
1.1	Background	1
1.2	Objective	3
2	Design	4
2.1	Problem Definition	4
2.1.1	Functional Requirements	4
2.1.2	Constraints	4
2.1.3	Criteria	4
2.2	CAD	4
3	Analysis	8
3.1	Flight Simulation	8
3.2	Loads Analysis	9
3.2.1	Flight Conditions	9
3.2.2	Fin Data	10
3.2.3	Body Profile Data	10
3.2.4	Loads Analysis Results	10
3.3	Longeron Failure Calculation	12
3.3.1	Total Loading on Individual Longeron	12
3.3.2	Failure Modes	13
3.3.3	Results	16
4	Conclusions and Next Steps	17
4.1	Conclusions	17
4.2	Next Steps	17
A	Aft Skirt Parameters	19

List of Figures

1	Labelled cross-section of KotS showing internal systems	1
2	Concept of Operations (CONOPS) diagram showing all stages of launch operations	2
3	Labelled cross-section of KotS showing internal systems	5
4	Labelled cross-section of the OTAS showing locations of fairings, launch tower, and fill port relative to the rest of the rocket	6
5	View of OTAS assembly showing curved tabs and fasteners	7
6	Openrocket GUI showing KotS	8
7	Axial loading at maxQ	11
8	Bending moment at maxQ	11
9	Axial loading at takeoff	12
10	Bending moment at takeoff	12

List of Tables

1	Total compressive load on longeron - maxQ	13
2	Total compressive load on longeron - Takeoff	13
3	Failure calculation inputs	14
4	Buckling failure - Geometric inputs	15
5	Calculated buckling failure loads	16
6	FOS for 3 failure modes	16

Summary

The purpose of this report is to document the design decisions and structural analysis of the oxidizer tank aft skirt (OTAS), a load-bearing assembly that is responsible for transmitting the thrust from the combustion chamber to the rest of the rocket. The CAD of the assembly is presented, and factors affecting the design such as ease of assembly and drag reduction are discussed. The structural analysis of the OTAS is described in 3 parts: Flight Simulation, Loads Analysis, and Failure Calculation. Flight simulation refers to the simulation of a high-level model of the entire rocket with the goal of predicting its trajectory from takeoff to ground hit. The flight simulation data and other aerodynamic, geometric, and inertial parameters are then used to evaluate the bending moments and axial loading across the entire rocket. Using the local bending moment and axial loading at the OTAS, the buckling, compressive, and tensile failure modes for the main structural component of the assembly are evaluated. Emphasis is placed upon the use of conservative assumptions whenever uncertainty exists at any stage of the analysis. The main structural component of the OTAS is found to have an acceptable safety factor, but further work must be done to ensure all components of the design are of an adequate structural integrity.

1 Introduction

1.1 Background

Waterloo Rocketry designs, manufactures, and launches a hybrid rocket each year as part of an international competition known as the Spaceport America Cup. Hybrid rockets use combustion reactants that are in different phases, most commonly liquid oxidizer and solid fuel, as is the case with the Waterloo Rocketry Team's Kismet engine. The rocket, named Kraken of the Sky (KotS) shown in Figure 1, is 6" in diameter and approximately 15' long and contains supporting electrical and parachute systems.

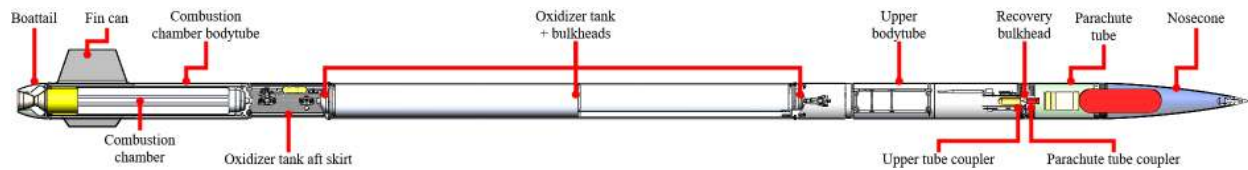


Figure 1: Labelled cross-section of KotS showing internal systems

The apogee, or maximum altitude target altitude that KotS is designed to reach is 30,000' Above Ground Level (AGL). Some of the factors that have a significant impact on apogee are weight, engine performance, and drag. The structure of the rocket must be capable of withstanding the loads that the rocket will experience through all flight regimes, as shown in 2. Sources of loading can be thrust from the engine, wind shear, and parachute deployment.

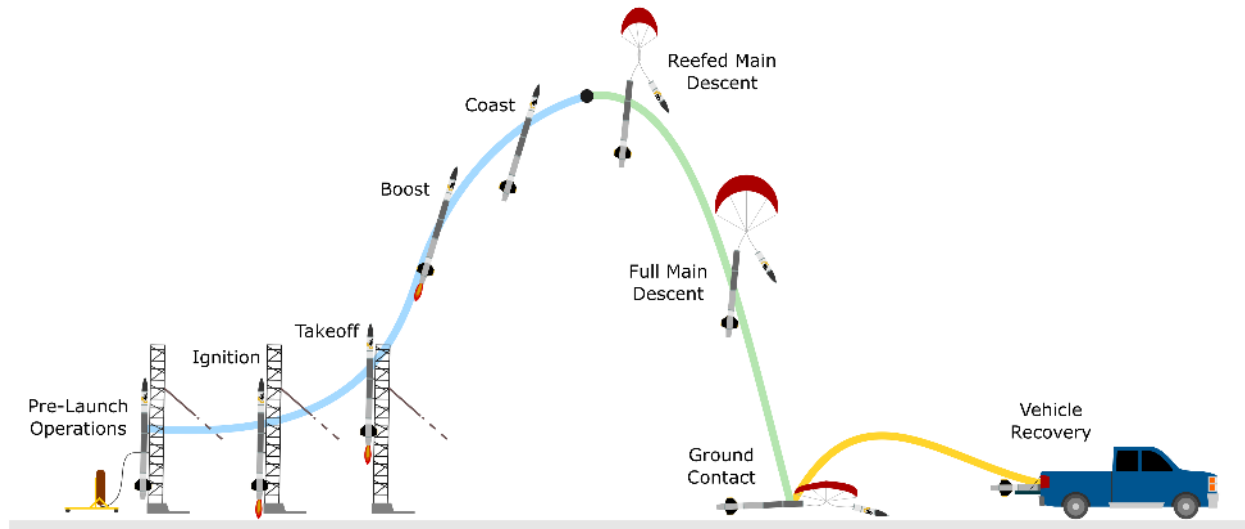


Figure 2: Concept of Operations (CONOPS) diagram showing all stages of launch operations

The structural assembly known as the Oxidizer Tank Aft Skirt (OTAS) is located between the combustion chamber, where the thrust from the combustion reaction between the propellants is generated, and the oxidizer tank, the pressure vessel containing the NOS. It is one of many systems contained in the injector section, which is defined by the space between the bottom of the oxidizer tank and the top of the combustion chamber. The other systems in the injector section are:

1. The injector valve and associated pneumatics, which opens to permit flow of oxidizer from the oxidizer tank to the combustion chamber
2. The fill plumbing, an assembly of high pressure fittings that includes a check valve that permits one-way NOS flow into the tank during fill operations, and a fill port that connects to the hose leading to the supply of NOS on the ground. The design of the hatch in the rocket that permits the ground plumbing to connect to the flight plumbing through the body of the rocket is outside of the scope of this report.
3. A section of braided hose connected in line with the injector valve that carries NOS into the combustion chamber when the injector valve opens
4. A custom PCB used for actuating the injector valve

All these systems are transported to the launch site and assembled onsite under a tight timeline, meaning that ease of assembly and access is critical to the success of the launch.

1.2 Objective

The previous OTAS design presented issues in terms of ease of assembly and access to internal components, so preliminary research was done to investigate existing solutions to this type of assembly. The preliminary research is described in [3]. In addition, the structural analysis performed on the old design was superficial and made use of excessively high factors of safety to account for the lack of in-depth structural analysis. The objective of this project is to redesign the OTAS to address the assembly issues and to better understand the loads experienced by the structure such that less conservative factors of safety can be used with more confidence.

2 Design

2.1 Problem Definition

The team has identified a need for an aft skirt design that is easy to assemble, permits easy access to internal systems, and minimizes weight and drag. Based off this needs statement, the following functional requirements, constraints, and criteria were defined.

2.1.1 Functional Requirements

1. Withstand all loads that the rocket will experience during launch operations without failure
2. Permit all required systems to fit within the space defined by the OTAS

2.1.2 Constraints

1. Outer diameter of 6"
2. Mechanically fastened connection to existing combustion chamber and oxidizer tank components

2.1.3 Criteria

1. Easy to assemble
2. Easy to manufacture
3. Light
4. Does not cause excessive drag

2.2 CAD

A circular array of 3 aluminum longerons was chosen as the design concept due to the ubiquity of longerons as axial load transferring structures in aircraft and rocket vehicles, with 3 being

the number that was determined to permit adequate space to access and assemble other components located in the injector section. The OTAS assembly in context with the rest of the rocket is shown in Figure 3. Aluminum 6061-T6 was chosen for its high strength to weight ratio and manufacturability.

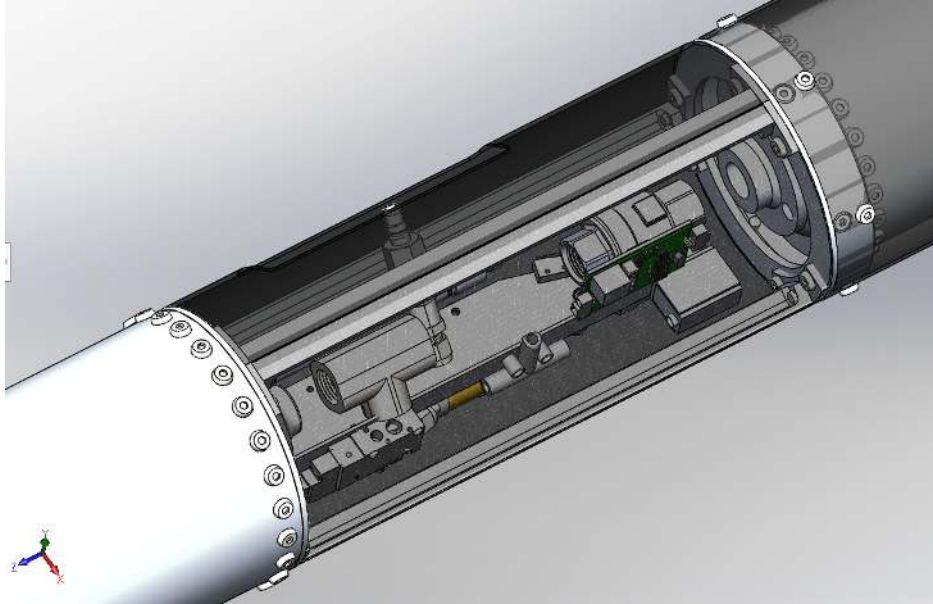


Figure 3: Labelled cross-section of KotS showing internal systems

A “T” shaped cross-section with flat ends on either end of the longerons was chosen in order to obtain a weight efficient area moment of inertia without the difficulties in fastening square or circular tubing to perpendicular surfaces without the use of welding, as well as to maximize internal area for the other systems in the injector section. These longerons are covered by 2 non structural fairings. One fairing covers an arc of 120° , and faces towards the launch rail. The other fairing covers an arc of 240° , and faces away from the launch rail such that it can be removed while the rocket is on the launch tower to allow access to the injector section internals. Due to the flexibility of the fairing panels, the large fairing can be bent enough to allow it to slip on and off the injector section without interference. This setup is illustrated in Figure 4.

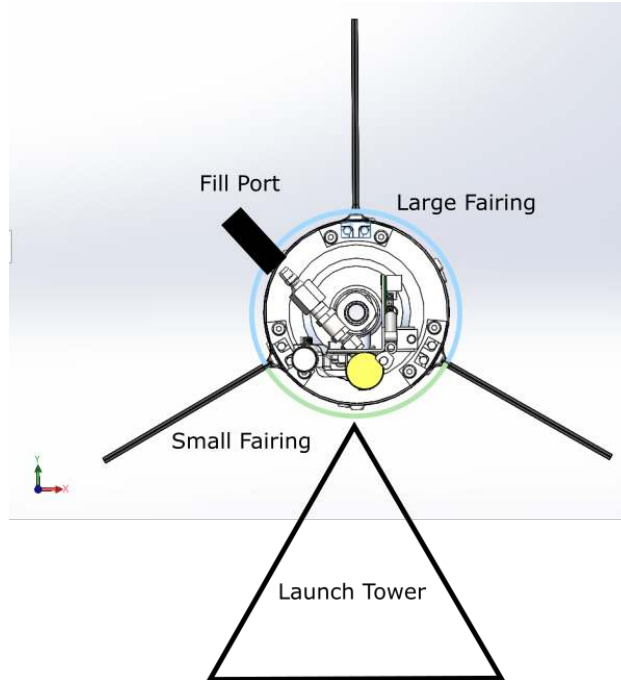


Figure 4: Labelled cross-section of the OTAS showing locations of fairings, launch tower, and fill port relative to the rest of the rocket

The geometry of the longerons is such that they can be manufactured using a manual mill. The ends of the longerons are symmetric, and are fastened to intermediate curved tabs, manufacturable on the waterjet cutter. Standard 87° countersunk 1/4-28 screws fasten the curved tabs to the longerons, and 1/4-28 button head cap screws fasten the curved tabs to the fill bulkhead located on the bottom of the oxidizer tank, and the coupler located on the injector bulkhead. These fasteners can be seen in Figure 5. The reason for selecting the 1/4-28 fastener size was due to the prevalence of these fasteners on the rest of the rocket. Using the same threading reduces the risk of an individual accidentally using a wrong size fastener and damaging the threads, while using common fasteners across different assemblies increases redundancy. If fasteners are lost or misplaced, it is more likely that there will be enough extra fastener stock to replace them.



Figure 5: View of OTAS assembly showing curved tabs and fasteners

In order to make the “guess-and-check” method of determining the dimensions of the longerons more efficient, all relevant design parameters such as the dimensions of the longeron cross-section, attachment points, and lengths are parametrically driven from the “Aft Skirt Parameters.txt” file in Appendix A. To update the dimensions, the new dimensions are entered into the text file. The next time the CAD model rebuilds, it automatically updates the dimensions that are linked to the text file.

3 Analysis

The following sections detail the process used to obtain the local forces experienced by the OTAS, starting with simulations of flight characteristics.

3.1 Flight Simulation

The team uses the open-source software Openrocket to simulate flight characteristics such as velocity, altitude, stability, and dynamic and aerodynamic coefficients from engine ignition to when the rocket comes back in contact with the ground. One of the key inputs to the flight simulation is the mass distribution of the rocket, which affects stability and apogee. In order to obtain estimates of masses and centers of gravity for every assembly on the rocket that are as close to the as-manufactured mass as possible, Solidworks mass properties are used as inputs to the Openrocket. Once flight components are manufactured and weighed, the Openrocket model is updated with the as-manufactured masses to further refine the simulation. Figure 6 shows KotS displayed in the Openrocket GUI.

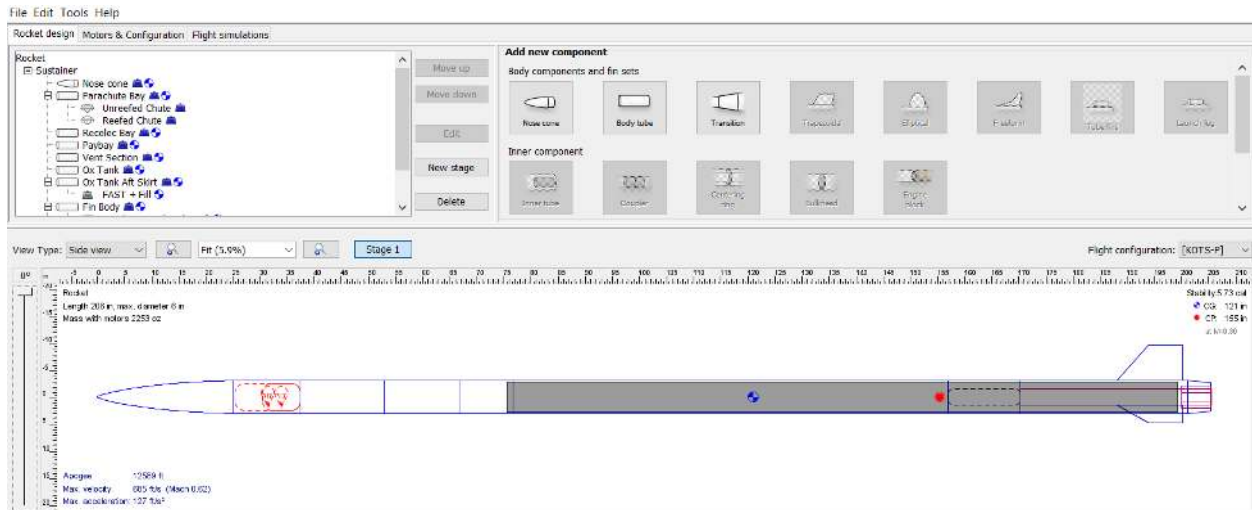


Figure 6: Openrocket GUI showing KotS

After the Openrocket model was updated with the mass distribution of the OTAS as measured in Solidworks, the simulated data was exported into a spreadsheet for further analysis.

3.2 Loads Analysis

In order to calculate the bending moments and axial loads across the length of the rocket, A spreadsheet and VBA based calculator developed for calculating loading on sounding rockets called “BENDIT7.xlsx” is used. The basic principles are described below. For more information, the equations used in and inputs to the calculator are discussed [9]. The BENDIT7 input data is primarily made up of 3 sections: Flight conditions, fin data, and body profile data. Appendix B contains the inputs used in BENDIT7 in order to perform this analysis.

3.2.1 Flight Conditions

BENDIT7 can only analyze loads at 1 point in time, so all time-varying flight conditions occur at the same timestamp in the Openrocket simulation. It is the user’s responsibility to ensure the time at which the flight conditions are chosen are representative of a worst case scenario. In this project, two notable flight regimes were analyzed: Takeoff, when axial acceleration is highest due to the thrust spike at the beginning of the burn that is characteristic of the Kismet engine, and maxQ, the time at which maximum dynamic pressure and therefore maximum drag occurs. MaxQ does not necessarily occur at the time corresponding to maximum velocity [6]. As shown in 1, drag is a function of both air density ρ and velocity v , where ρ decreases as altitude increases. The times at which the analysis was performed were found using basic spreadsheet equations to be $t = 4.6\text{s}$ for maximum axial acceleration and $t = 15.341\text{s}$ for maxQ.

$$F = \frac{1}{2}\rho v^2 \tag{1}$$

Most of the inputs are found in either the exported Openrocket simulation data or other information contained in the Openrocket model. Some inputs, such as thrust offset, are unknown for the Kismet engine and are also not critical to the calculation of loads on the rocket so long as the user is aware of the effects that these inputs can have on the final safety

factors. In these cases, values of an equal or greater conservatism than the sample BENDIT7 inputs were used. The inputs used in the analysis of the OTAS can be found in Appendix B.

3.2.2 Fin Data

The fin data is for the most part taken straight from the existing fin geometry or obtained from the Openrocket Component Analysis tool. In some cases, the values for some inputs were unknown so the original values provided in the sample implementation of the BENDIT7 calculator were retained. These values are: $1 \sigma \delta$ (fin), and Single Panel $C_{N\delta}$.

3.2.3 Body Profile Data

The body profile data requires the user to decide how to section the rocket into “elements” such that the mass of each element is roughly centered along the length of the element. For simplicity, in this particular implementation of BENDIT7, the start and end of each element corresponds to the start and end of a section in Openrocket. BENDIT7 has the capability to perform more detailed analysis of some stochastic characteristics, but these were not used in the evaluation of the loading on the KotS rocket.

3.2.4 Loads Analysis Results

As expected, at maxQ the bending moments are more significant than axial force, while at takeoff axial force is more significant than the bending moments. This is because the bending moments are induced by aerodynamic forces caused by gust loading as the rocket passes through layers of wind moving in different directions, and gust amplitude increases at higher altitudes. The bending moment is at a maximum at body station 157, measured from the tip of the rocket, which is coincident with the top of the injector section, and the axial force is at a maximum at body station 170, coincident with the bottom of the injector section. The location of maximum moment and force holds for both flight regimes that were investigated. The moment and axial loads used in the determination of total axial compression experienced by the longerons are the maximum of the values found between

body station 157 - 170, which is the more conservative assumption. The full bending and axial load diagrams are shown in Figures 7, 8, 9, and 10.

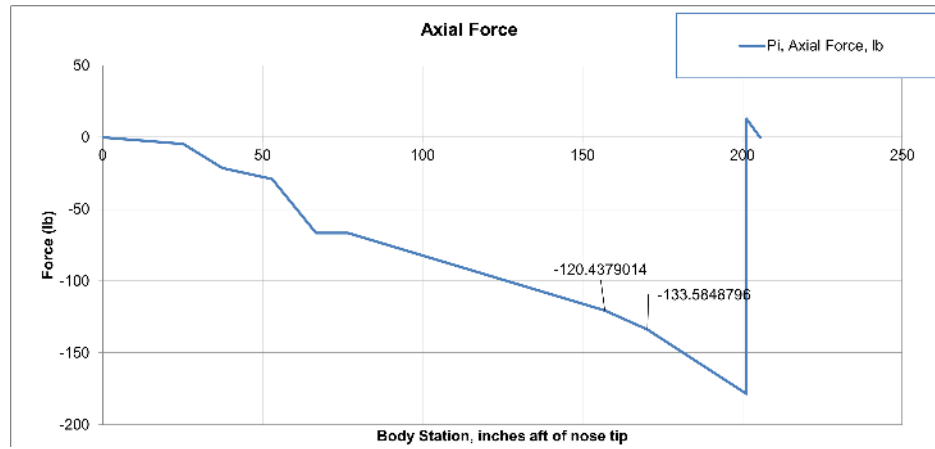


Figure 7: Axial loading at maxQ

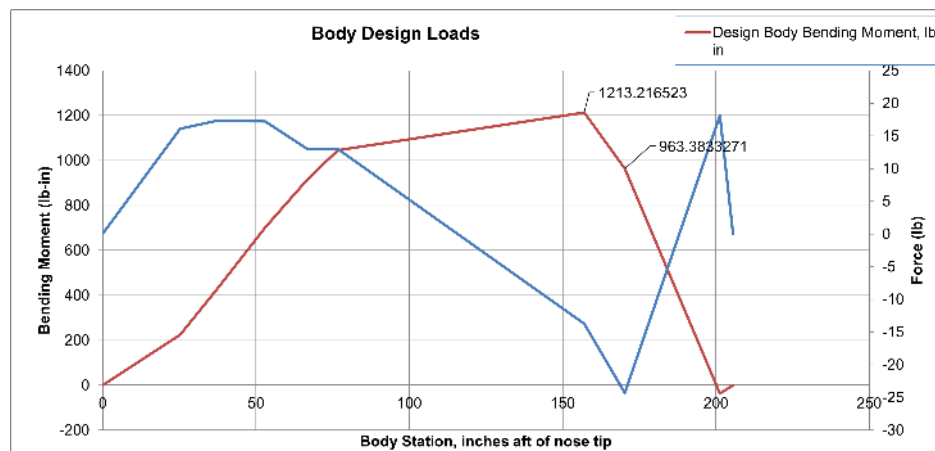


Figure 8: Bending moment at maxQ

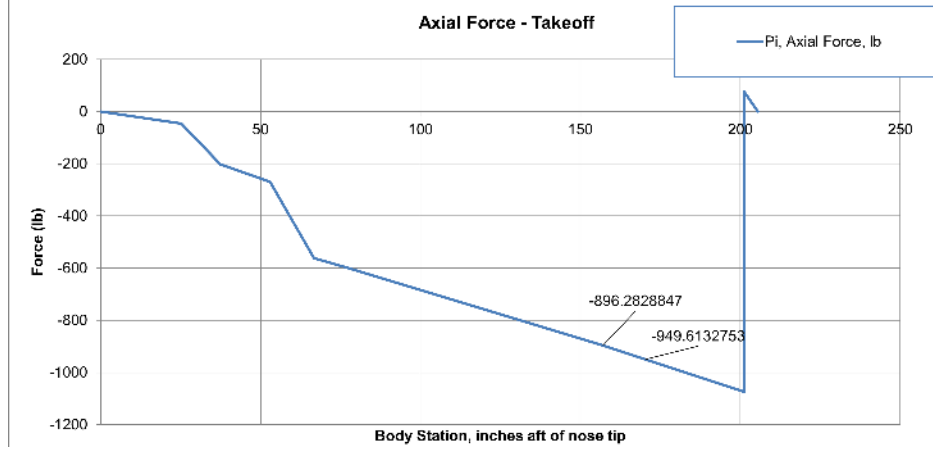


Figure 9: Axial loading at takeoff

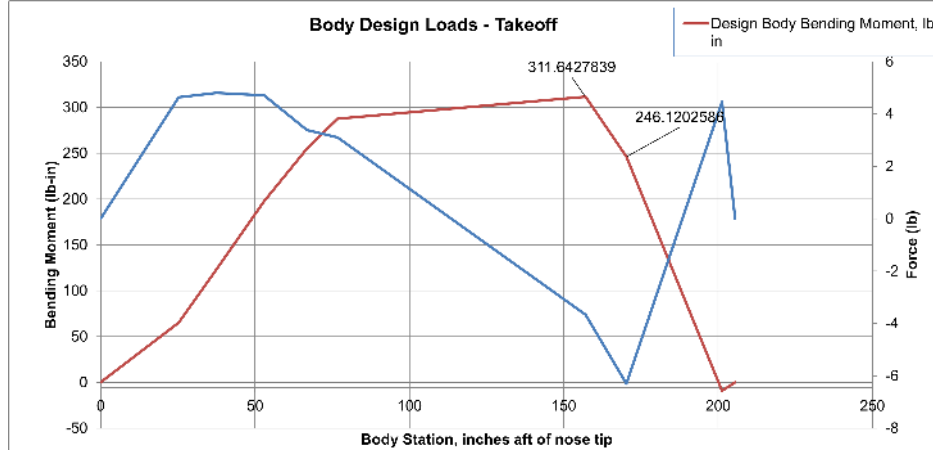


Figure 10: Bending moment at takeoff

3.3 Longerons Failure Calculation

3.3.1 Total Loading on Individual Longerons

In order to determine the total loading experienced by each individual longeron P_{max} due to bending of the overall structure, Equation 2 derived in [10] is used. This equation is valid for longerons arrayed at equidistant angular positions about the centerline of the vehicle, and assumes that the bending deflection is negligible, such that planes that are parallel to the cross-sections of the longerons stay parallel to each other after bending. The total load on the longeron P_{tot} is then the sum of the axial load P and P_{max} . The inputs to the calculation

of total loading on an individual longeron are summarized in Tables 1 and 2 for maxQ and takeoff respectively. It was found that the worst compressive loading is 392psi and occurs at $t = 4.6s$ at takeoff.

$$P_{tot} = \pm \frac{2M}{RN} \quad (2)$$

Table 1: Total compressive load on longeron - maxQ

Item	Description	Value	Unit	Source
M	Moment at STA 157	1211	lb-in	BENDIT7
P	Axial Load at STA 170	135	lbf	BENDIT7
R	Distance - centroid to centerline	3	in	SW Properties
N	Number of longerons	3	unitless	Constraint
P_{max}	Maximum load on longeron due to bending	293	lbf	Calculated
P_{tot}	Total load on longeron	338	lbf	Calculated

Table 2: Total compressive load on longeron - Takeoff

Item	Description	Value	Unit	Source
M	Moment at STA 157	311	lb-in	BENDIT7
P	Axial Load at STA 170	950	lbf	BENDIT7
R	Distance - centroid to centerline	3	in	SW Properties
N	Number of longerons	3	unitless	Constraint
P_{max}	Maximum load on longeron due to bending	75	lbf	Calculated
P_{tot}	Total load on longeron	392	lbf	Calculated

3.3.2 Failure Modes

3 different failure modes for the longeron were evaluated: Buckling, compressive, and tensile. The inputs used in the evaluation of these failure modes are summarized in Table 3.

Table 3: Failure calculation inputs

Item	Description	Value	Unit	Source
C	End Condition (pinned-pinned)	1	unitless	Conservative assumption
E	Young's Modulus	10	Mpsi	[8]
σ_y	Yield Strength	40	ksi	[8]
l	length of longeron	13.00	in	Constraint
N	Number of longerons	3	unitless	Constraint
F_d	Deployment load on longeron	336	lbf	Constraint
P_{tot}	Total load on longeron	392	lbf	Calculated

Buckling failure of columns is governed by different equations depending on the slenderness ratio $\frac{l}{k}$ of the column, calculated as the ratio between the length of the column and the radius of gyration k calculated in Equation 3. If $\frac{l}{k}$ is greater than the slenderness ratio limit $\frac{l}{k_l}$ calculated in Equation 4, then the column can be considered to be a slender column rather than an intermediate column [5]. The longeron fulfills this criterion, so the buckling failure equation used is for that of a slender column. The geometric inputs to the column failure calculation and calculated values for determination of column type are summarized in 4.

$$k = \sqrt{\frac{I}{A}} \quad (3)$$

$$\left(\frac{l}{k}\right)_l = \left(\frac{2\pi^2 CE}{S_y}\right)^{0.5} \quad (4)$$

Table 4: Buckling failure - Geometric inputs

Item	Description	Value	Unit	Source
c	Distance to N.A.	0.375	in	Design Parameter
I_x	Moment of inertia about x	0.0037	in ⁴	SW Properties
A	Area	0.1875	in ²	SW Properties
k	Radius of gyration about x	0.1409	in	Calculated
$\frac{l}{k}$	Slenderness ratio	92	unitless	Calculated
$\frac{l}{k_l}$	Slenderness ratio limit	70	unitless	Calculated

In the determination of the maximum compressive loading on the individual longerons, it was assumed that the longerons experience no bending, which may not necessarily be a safe assumption. To account for this, the buckling failure calculation assumes a conservative eccentricity ratio e_r of 1 [4], which results in a load offset of 0.0529" calculated in Equation 5 by rearranging and solving for e . This means that the failure calculation accounts for manufacturing defects and structural loading that causes the load to be offset from the centroid of the cross-section by 0.0529", inducing a bending moment in the longeron.

$$e_r = \frac{ec}{k^2} \quad (5)$$

The equation for the maximum stress σ_{max} experienced by a column in eccentric axial loading is given in Equation 6, where P_{crit} is the Euler buckling load, which is the load at which the column would buckle if there were zero eccentricity [1], calculated using Equation 7.

$$\sigma_{max} = P \left[\frac{1}{A} + \frac{ec}{I} \sec \left(\frac{\pi}{2} \sqrt{\frac{P}{P_{cr}}} \right) \right] \quad (6)$$

$$P_{cr} = \frac{\pi^2 EI}{L^2} \quad (7)$$

A guess and check process was used to identify P , the load at which σ_{max} equals the yield stress of the material σ_y . The safety factor for buckling failure is then the ratio between P

and P_{tot} , and was found to be 3.86. Table 5 summarizes the values found in the buckling failure calculations.

Table 5: Calculated buckling failure loads

Item	Description	Value	Unit	Source
P_{cr}	Euler critical buckling load	2172	lbf	Calculated
e	Eccentric offset	0.0529	in	Eccentricity ratio = 1 [4]
σ_{max}	Maximum stress in longeron	39622	psi	Calculated
P	Eccentric buckling load guess	1515	lbf	Iterated Value
FOS_b	Safety factor-eccentric buckling	3.86	unitless	Calculated

For the 2nd and 3rd failure modes, compressive and tensile failure respectively, the stresses are calculated by dividing the given load by the longeron cross-sectional area. The compressive load was taken to be P_{tot} , and the tensile load was taken to be F_d , the shock loading experienced by the rocket during parachute deployment. The safety factors are 4.95 and 22.30 in compression and tension respectively.

3.3.3 Results

The safety factors for the longeron failure modes are summarized in Table 6. It can be seen that eccentric buckling is the limiting failure mode, but is an acceptable value.

Table 6: FOS for 3 failure modes

Item	Description	Value	Unit	Source
FOS_e	Safety factor - eccentric buckling	3.86	unitless	Calculated
FOS_c	Safety factor - compressive yielding	4.95	unitless	Calculated
FOS_t	Safety factor - tensile yielding	22.30	unitless	Calculated

4 Conclusions and Next Steps

4.1 Conclusions

The design of systems on a hybrid sounding rocket for a student design team often involves interesting problems, and a wealth of learning opportunity in topics not covered in the typical mechanical engineering curriculum. In this case, the drawback is that there is more uncertainty regarding the trustworthiness of calculations performed in the analysis of the OTAS, leading to the application of conservative assumptions. This analysis involved the use of multiple tools in order to obtain the intermediate values to be used in the next stage of the calculations: The Openrocket simulation software, spreadsheets, and a spreadsheet and VBA based calculator. Using these tools, the strength of the longerons was found to be adequate.

4.2 Next Steps

The process detailed in this report involves many steps. Mistakes made at any point in the analysis could result in an overly optimistic result, which could have catastrophic consequences when the structure of the rocket is put to the test during launch. To mitigate analysis mistakes, a review process by other team members and advisors knowledgeable in loads and structural analysis is critical. Training for team members and an emphasis on an understanding of rocket aerodynamics and structural analysis is recommended.

Notably omitted from this report is a discussion of the strength of the fastened joints. In order for the analysis of the OTAS to be considered complete, the following considerations must be made:

1. Thread shear-out of the button-head and countersunk 1/4-28 fasteners on the top and bottom of the OTAS, a total of 4 failure calculations
2. Tensile failure of the aforementioned fasteners
3. Preload required to prevent joint separation for the aforementioned fasteners

4. Combined loading failure of the curved tabs on either end of the longerons connecting them to the bulkheads, most likely requiring the use of FEA
5. Combined loading failure of the injector bulkhead coupler, most likely requiring the use of FEA

A Aft Skirt Parameters

"Injector Section Length"=13in

"Injector Leg Thiccnness"=0.125in

"Injector Leg Width From Arm"=0.1875in

"Injector Leg Height From Web"=0in

"Injector Leg Fillet 1"=0.0625in

"Injector Leg Overall Width"=2*"Injector Leg Width From Arm"+2*"T Arm Height"
+"T Web Thiccnness"

"Injector Leg Overall Height"="T Arm Thiccnness"+"T Web Height"
+"Injector Leg Height From Web"

"T Web Thiccnness"=0.1875in

"T Web Height"=0.375in

"T Arm Thiccnness"=0.125in

"T Arm Height"=0.375in

"Inj-Hole Distance From Web"="Injector Leg Fillet 1"+0.2625in

"Fill-Hole Distance From Web"=0.2625in

"Fill-Hole Distance From Bulkhead"=0.2625in

"Fairing-Hole Angle From Centerline"=162.425in

"Fairing-Hole Edge Distance"=0.11in

"Fill-Tab Height"=0.5in

"Shim Thiccnness"=0.0625in

"Hatch Distance From Bulkhead"=2in

"Hatch Square Width"=2in

"Fairing Screw OD"=0.138in

"Fairing Thiccnness"=0.0625in

References

- [1] B. McGinty, "Buckling of eccentrically loaded columns," Eccentric Column Buckling. [Online]. Available: <https://www.continuummechanics.org/eccentriccolumnbuckling.html>. [Accessed: 20-Sep-2021].
- [2] H. Abu Abah, Z. Bian, J. Deery, D. Dymment, T. Tang, A. Hussien, D. Kletke, R. Kobets, S. Kong, A. Leszkowiat, Z. Mears, C. Sankey, A. Sotnikov, K. Tam, L. Tomlin, and W. Yu, "Design Proposal Report for Kraken of the Sky Competing Team 6," Jan. 2021
- [3] [1]S. Kong, "Preliminary Design and Analysis of a Hybrid Rocket Structure", Waterloo, 2021.
- [4] "Column Buckling — MechaniCalc", Mechanicalc.com, 2021. [Online]. Available: <https://mechanicalc.com/reference/column-buckling>. [Accessed: 20- Sep- 2021].
- [5] Budynas, Shigleys, "*Mechanical Engineering Design*", 11th Edition, Si Units. S.L.: McGraw-Hill Education. 2021-04-12
- [6] aaronridley, "Max-Q and Why Does it Matter?", The Rocket Science Blog, 2021. [Online]. Available: <https://therocketscienceblog.wordpress.com/2018/06/24/max-q-and-why-does-it-matter/>. [Accessed: 20- Sep- 2021].
- [7] S. Niskanen, "Openrocket technical documentation", 2013.
- [8] "ASM Material Data Sheet", Asm.matweb.com. [Online]. Available: <http://asm.matweb.com/search/SpecificMaterial.asp?bassnum=MA6061T6>. [Accessed: 20- Sep- 2021].
- [9] C. Hoult, "Sounding Rocket Structural Design Loads (rev.9)", Culver City, 2014.
- [10] C. Hoult, "Design Bending Loads for a Multiple Longeron Structure", Culver City, 2013.